

Conditions for Performing Inference

- Random
 - Two random samples (Sampling)
 - Randomly Assigned to two groups (Experiment)
- Independent
 - Sampling – Check 10% condition on BOTH
 - Experiment – Condition doesn't need to be checked
- Normal/Large Sample
 - Both populations are Normally Distributed
 - Both sample sizes are larger than 30 (CLT)
 - If one or both are smaller than 30, graph to check for no outliers or strong skewness

Two Sample T Test for Population Means

When the conditions are met and $H_0: \mu_1 = \mu_2$:

- Find the P-value by calculating the probability of getting a t statistic this large or larger in the direction specified by the alternative hypothesis H_a .
- Use the t distribution with degrees of freedom approximated by the smaller of $n_1 - 1$ and $n_2 - 1$ OR the degrees of freedom given by technology

$$H_a: \mu_1 > \mu_2$$

$$H_a: \mu_1 \neq \mu_2$$

$$H_a: \mu_1 < \mu_2$$

Example: Two Sample T Test for a Difference in Means

Does increasing the amount of calcium in our diet reduce blood pressure? Examination of a large sample of people revealed a relationship between calcium intake and blood pressure, and the relationship was strongest for black men. Because observational studies do not allow for us to draw a cause-and-effect conclusion, a randomized experiment was designed. The subjects were 21 healthy black men who volunteered to take part in the double-blind experiment. They were randomly assigned to two groups: 10 of the men received a calcium supplement for 12 weeks, while the control group of 11 men received a placebo pill that looked identical. The response variable is the decrease in systolic (top number) blood pressure for a subject after 12 weeks, in millimeters of mercury. An increase appears as a negative number. Here are the data:

Group 1 (calcium)	7	-4	18	17	-3	-5	1	10	11	-2	
Group 2 (placebo)	-1	12	-1	-3	3	-5	5	2	-11	-1	-3

Do the data provide convincing evidence that a calcium supplement reduces blood pressure more than a placebo? Carry out an appropriate test to support your answer then interpret the P-value you got in the context of this experiment.